# Manufacturing Processes Lab (23A03401P)

II Year B.Tech ME, II Semester | JNTUA R23

**Course code:** 23A03401P

**Credits:** L = 0, T = 0, P = 3, C = 1.5

**Course Objective:** Acquire practical knowledge on Metal Casting, Welding, Press Working and Processing of Plastics.

## Course Outcomes

| CO | Statement | Blooms |
|---|---|---|
| CO1 | Make moulds for sand casting. | L2 |
| CO2 | Fabricate different types of components using various manufacturing techniques. | L5 |
| CO3 | Adapt unconventional manufacturing methods. | L3 |
| CO4 | Develop different weld joints. | L6 |
| CO5 | Explain different types of 3D printing techniques. | L2 |

Related COs shown for each experiment are suggested by topic.

**Each experiment lists: Related CO, Aim, Apparatus, INPUTS (what you set / measure), Procedure, OUTPUTS (what you calculate / get).**

## List of Experiments

1. Design and making of pattern (i. single piece, ii. split)
2. Sand properties testing: (i) Sieve analysis (dry sand)
3. Sand properties testing: (ii) Clay content test
4. Sand properties testing: (iii) Moisture content test
5. Sand properties testing: (iv) Strength test (compression and shear)
6. Sand properties testing: (v) Permeability test
7. Mould preparation (straight pipe, bent pipe, dumble, gear blank)
8. Gas cutting and welding
9. Manual metal arc welding (i. lap joint, ii. butt joint)
10. Injection moulding
11. Blow moulding
12. Simple models using sheet metal operations
13. Study of deep drawing and extrusion operations
14. To make weldments using TIG / MIG welding
15. To weld using spot welding machine
16. To join using brazing and soldering
17. To make simple parts on a 3D printing machine
18. Demonstration of metal casting

**Virtual lab:** 5 simulations (at the end)

## 1. Design and making of pattern (i. single piece, ii. split)

**Related CO:** CO1

**Aim:** To design and make a single piece pattern and a split pattern for a given casting.

**Apparatus:** Wood (teak/pine), saw, plane, chisel, mallet, steel rule, try square, marking gauge, sand paper.

**INPUTS:** Component drawing; pattern material; allowances: shrinkage (about 1 percent for CI), machining, draft (1 to 3 degrees), shake.

**Procedure:**

1) Study the drawing and select the parting line.
2) Add shrinkage, machining and draft allowances to the dimensions.
3) Mark, cut, plane and finish the wood to size.
4) For split pattern make two halves and join them with dowel pins.
5) Apply shellac and colour code the pattern.

**OUTPUTS:** A finished single piece pattern and a two-half split pattern with dowel pins, core prints and draft.

**Precautions:** Keep tools sharp and clamp the work firmly.

## 2. Sand properties testing: (i) Sieve analysis (dry sand)

**Related CO:** CO1

**Aim:** To find the grain fineness number (GFN) of the base sand.

**Apparatus:** Sieve shaker, standard sieves, balance, dry sand.

**INPUTS:** 50 g of clean dry sand; sieve set (mesh numbers 6 to 300 and pan); shaking time about 15 min.

**Procedure:**

1) Weigh 50 g of dry sand.
2) Stack sieves with coarse on top and pan at the bottom.
3) Shake for 15 minutes.
4) Weigh sand retained on each sieve.

**OUTPUTS:** Percentage retained on each sieve and the AFS grain fineness number (GFN).

**Formulae:** GFN = (sum of W x M) / (sum of W), where W = weight retained and M = multiplying factor of the sieve.

## 3. Sand properties testing: (ii) Clay content test

**Related CO:** CO1

**Aim:** To find the percentage of clay in the moulding sand.

**Apparatus:** Clay content tester (wash bottle), caustic soda 3 percent, distilled water, drying oven, balance.

**INPUTS:** 50 g of dried sand; 475 ml distilled water + 25 ml of 3 percent NaOH; stirring 5 min; settling 10 min (repeat till water is clear).

**Procedure:**

1) Weigh 50 g of dry sand.
2) Add the NaOH water solution and stir for 5 minutes.
3) Let it settle, siphon out the cloudy water, repeat until clear.
4) Dry the remaining sand and weigh it.

**OUTPUTS:** Percentage of clay content of the sample.

**Formulae:** Clay % = (W1 - W2) / W1 x 100; W1 = initial weight, W2 = weight after washing and drying.

## 4. Sand properties testing: (iii) Moisture content test

**Related CO:** CO4

**Aim:** To find the moisture percentage in the moulding sand.

**Apparatus:** Moisture teller or oven, balance, sample tray.

**INPUTS:** 20 to 50 g of green sand; drying temperature 105 to 110 deg C; drying time about 1 hr.

**Procedure:**

1) Weigh a green sand sample (W1).
2) Dry it in the oven at 105 to 110 deg C.
3) Cool and weigh the dried sample (W2).

**OUTPUTS:** Percentage moisture in the sand (normally 2 to 6 percent for green sand).

**Formulae:** Moisture % = (W1 - W2) / W1 x 100

## 5. Sand properties testing: (iv) Strength test (compression and shear)

**Related CO:** CO4

**Aim:** To find the green compressive strength and shear strength of the moulding sand.

**Apparatus:** Universal sand strength testing machine, sand rammer, standard specimen tube (50 mm dia x 50 mm height).

**INPUTS:** Sand mix with known moisture; 3 rams of the rammer for the standard specimen; load applied slowly by hand wheel.

**Procedure:**

1) Prepare the standard specimen with 3 rammer blows.
2) Fix it between the compression pads and apply load until failure.
3) Note the dial reading in N/cm2 (or kg/cm2).
4) Repeat with shear fixtures for shear strength.

**OUTPUTS:** Compressive strength (about 5 to 12 N/cm2 for green sand) and shear strength (about 1 to 3 N/cm2). Plot strength vs moisture if several moisture levels are tested.

**Formulae:** Strength = Load at failure / Area of cross section of the specimen

## 6. Sand properties testing: (v) Permeability test

**Related CO:** CO2

**Aim:** To find the permeability number of the moulding sand.

**Apparatus:** Permeability meter, standard rammer, specimen tube, stop watch.

**INPUTS:** Standard specimen (50 mm dia x 50 mm); air volume V = 2000 cc; pressure p (g/cm2) from the manometer; time t (min).

**Procedure:**

1) Prepare a standard specimen.
2) Fix the tube on the permeability meter.
3) Pass 2000 cc air through the sand and note time and pressure.

**OUTPUTS:** Permeability number of the sand. Higher value means more venting capacity.

**Formulae:** $P = (V \times h) / (p \times A \times t)$; h = specimen height (5.08 cm), A = area (20.268 cm2).

## 7. Mould preparation (straight pipe, bent pipe, dumble, gear blank)

**Related CO:** CO2

**Aim:** To prepare sand moulds for the given patterns.

**Apparatus:** Moulding boxes (cope and drag), pattern, moulding sand, parting sand, rammer, sprue pin, riser pin, trowel, vent wire, gate cutter.

**INPUTS:** Pattern (solid, split or with core prints); green sand; core for hollow pipe; gating details.

**Procedure:**

1) Place the drag with the pattern half on a board and sprinkle parting sand.
2) Fill with sand and ram it; turn the drag over.
3) Place the cope, sprue pin and riser pin; ram the sand.
4) Remove the pins and cope, then draw the pattern.
5) Cut gates, vent the mould, place the core if needed, and close the box.

**OUTPUTS:** Ready moulds for (i) straight pipe, (ii) bent pipe, (iii) dumble, (iv) gear blank, ready for pouring.

**Precautions:** Avoid over ramming; vent properly to prevent blow holes.

## 8. Gas cutting and welding

**Related CO:** CO2

**Aim:** To cut and weld mild steel plates using oxy-acetylene gas.

**Apparatus:** Oxygen and acetylene cylinders with regulators, torch, hoses, cutting torch, flint lighter, goggles, MS plates.

**INPUTS:** MS plate (2 to 5 mm); gas pressures: oxygen 1 to 1.5 bar, acetylene 0.15 bar for welding; neutral flame; filler rod.

**Procedure:**

1) Set the regulators and light acetylene first, then add oxygen to get a neutral flame.
2) For cutting: preheat the edge to red hot and press the cutting oxygen lever.
3) For welding: melt the edges, make a pool and feed the filler rod.

**OUTPUTS:** Cut plate pieces and a welded butt joint; visual check for penetration and bead quality.

**Precautions:** Wear goggles, check leaks, light acetylene before oxygen.

## 9. Manual metal arc welding (i. lap joint, ii. butt joint)

**Related CO:** CO2

**Aim:** To make a lap joint and a butt joint using shielded metal arc welding.

**Apparatus:** AC/DC welding machine, electrode holder, earth clamp, flux-coated electrodes (E6013), MS plates, chipping hammer, wire brush, helmet, gloves.

**INPUTS:** MS plates; current 80 to 150 A (depends on plate and electrode dia); voltage about 20 to 30 V; polarity; joint set up.

**Procedure:**

1) Clean the plates and set them in lap or butt position.
2) Tack weld both ends.
3) Strike the arc and run the bead with uniform speed.
4) Chip off slag and brush.

**OUTPUTS:** Lap joint and butt joint specimens; inspect bead appearance, penetration, and defects.

**Precautions:** Use a helmet and gloves; never look at the arc directly.

## 10. Injection moulding

**Related CO:** CO3, CO4

**Aim:** To make plastic parts by injection moulding.

**Apparatus:** Injection moulding machine, mould/die, hopper, plastic granules (PP or HDPE).

**INPUTS:** Granule material; barrel temperature (about 180 to 230 deg C for PP); injection pressure; clamp force; cooling time.

**Procedure:**

1) Close the mould and feed granules into the hopper.
2) Granules melt in the heated barrel and the screw injects them into the mould.
3) Hold pressure, cool, open the mould and eject the part.

**OUTPUTS:** Moulded plastic component; observe defects like flash, short shot, sink marks.

## 11. Blow moulding

**Related CO:** CO3, CO4

**Aim:** To make hollow plastic articles (bottles) by blow moulding.

**Apparatus:** Blow moulding machine, extruder, split mould, compressed air supply, HDPE/PET granules.

**INPUTS:** Plastic granules; melt temperature; air pressure (about 3 to 6 bar); mould cooling time.

**Procedure:**

1) Extrude a hot hollow tube (parison).
2) Close the split mould around the parison.
3) Blow compressed air so the plastic takes the shape of the mould.
4) Cool, open and remove the bottle; trim flash.

**OUTPUTS:** Hollow plastic bottle; check wall thickness and finish.

## 12. Simple models using sheet metal operations

**Related CO:** CO2, CO4

**Aim:** To make simple models (tray, funnel, box) using sheet metal operations.

**Apparatus:** GI sheet, shears, snips, steel rule, scriber, divider, mallet, stakes, hand folding tools, soldering iron.

**INPUTS:** Sheet thickness; development (layout) of the model; allowances for seams and hems.

**Procedure:**

1) Draw the development on the sheet.
2) Cut with snips (shearing).
3) Bend and fold on the stakes.
4) Join with seams or by soldering.

**OUTPUTS:** Finished sheet metal model (tray or funnel).

**Precautions:** Deburr sharp edges; wear gloves.

## 13. Study of deep drawing and extrusion operations

**Related CO:** CO3, CO5

**Aim:** To study deep drawing of a cup and extrusion process.

**Apparatus:** Hydraulic or mechanical press, punch, die, blank holder, lubricant, sheet blank; extrusion setup (billet, container, die).

**INPUTS:** Blank diameter and thickness; punch and die dimensions; blank holder force; drawing speed; billet material and temperature.

**Procedure:**

1) Place the lubricated blank on the die.
2) Hold it with the blank holder and push down the punch.
3) For extrusion, push the billet through the die with a ram.

**OUTPUTS:** Drawn cup (check drawing ratio, wrinkling, tearing) and extruded section of the die profile.

**Formulae:** Drawing ratio = Blank diameter / Punch diameter

## 14. To make weldments using TIG / MIG welding

**Related CO:** CO1

**Aim:** To make weld joints using TIG and MIG welding.

**Apparatus:** TIG and MIG welding machines, argon gas cylinder, tungsten electrode, filler wire or rod, regulator, plates.

**INPUTS:** Plate material and thickness; current/voltage; shielding gas flow (about 8 to 15 l/min); wire feed speed (MIG); travel speed.

**Procedure:**

1) Clean and fix the plates.
2) Set the current, gas flow and wire feed.
3) Weld with a steady torch angle and speed.
4) Inspect the bead.

**OUTPUTS:** Clean, spatter-free weldments in TIG (no slag) and MIG (high deposition) joints.

**Precautions:** Use a helmet; ensure good ventilation.

## 15. To weld using spot welding machine

**Aim:** To join thin sheets by resistance spot welding.

**Apparatus:** Spot welding machine, copper electrodes, MS sheets, cooling water.

**INPUTS:** Sheet thickness; welding current; electrode force; weld time (cycles).

**Procedure:**

1) Overlap and clean the sheets.
2) Place between electrodes.
3) Press the pedal; current flows and a nugget forms.
4) Release after the hold time.

**OUTPUTS:** Spot welded lap joint with a weld nugget; check by peel/tear test.

## 16. To join using brazing and soldering

**Aim:** To join metals by brazing (high temperature filler) and soldering (low temperature filler).

**Apparatus:** Gas torch, soldering iron, brass or silver filler, solder (Pb-Sn or lead free), flux (borax, rosin), copper/MS pieces.

**INPUTS:** Base metals; filler metal; flux type; heating temperature (soldering below 450 deg C, brazing above 450 deg C).

**Procedure:**

1) Clean the joint and apply flux.
2) Heat the joint (not the filler).
3) Touch the filler so it flows by capillary action.
4) Cool and clean flux residue.

**OUTPUTS:** Soldered joint (electrical type) and brazed joint (stronger).

## 17. To make simple parts on a 3D printing machine

**Aim:** To print simple parts using a 3D printer (FDM).

**Apparatus:** FDM 3D printer, PLA/ABS filament, computer with CAD and slicer software.

**INPUTS:** CAD model (STL file); slicer settings: layer height (0.1 to 0.3 mm), infill %, print speed, nozzle (about 200 deg C) and bed temp, supports.

**Procedure:**

1) Design a part in CAD and export as STL.
2) Slice it and generate G-code.
3) Level the bed and load the filament.
4) Print, cool and remove the part; remove supports.

**OUTPUTS:** A printed part; check dimensional accuracy, layer quality, and print time.

## 18. Demonstration of metal casting

**Aim:** To demonstrate the sand casting process.

**Apparatus:** Furnace (crucible/induction), ladle, mould, molten metal (aluminium), tongs, safety wear.

**INPUTS:** Metal charge; pouring temperature (aluminium about 700 to 750 deg C); prepared mould.

**Procedure:**

1) Melt the metal in the furnace.
2) Preheat the ladle and skim the slag.
3) Pour into the mould at the correct temperature.
4) Let it solidify, then knock out and fettle the casting.

**OUTPUTS:** Solid casting; inspect for defects such as blow holes, shrinkage cavities and misrun.

**Precautions:** Wear leather apron, gloves, face shield; keep moulds dry.

## Virtual Lab

### 1. Stages of casting (sand casting process)

**Link:** https://virtual-labs.github.io/exp-sand-casting-process-dei/theory.html

**INPUTS:** Choose pattern, mould type, metal and pouring conditions in the simulation.

**OUTPUTS:** Visual of each casting stage: pattern, moulding, melting, pouring, solidification, finishing.

### 2. Oxy-acetylene welding and cutting setup

**Link:** https://virtual-labs.github.io/exp-gas-cutting-processes-iitkgp/index.html

**INPUTS:** Gas pressures, flame type (neutral, carburising, oxidising), torch angle, cutting speed.

**OUTPUTS:** Cut or welded plate; observe kerf quality and flame effect.

### 3. Simulation of Fused Deposition Modelling (FDM)

**Link:** https://3dpdei.vlabs.ac.in/exp/simulation-modelling-process

**INPUTS:** Layer thickness, infill, nozzle temperature, speed and part orientation.

**OUTPUTS:** Simulated printed part, print time, material used.

### 4. Altair Inspire Mould

**Link:** https://altair.com/inspire-mold/

**INPUTS:** CAD part, gate and runner position, material, process parameters.

**OUTPUTS:** Mould filling and defect prediction (short shot, air traps, weld lines).

### 5. Simulation of Cartesian system (3D printer motion)

**Link:** https://virtual-labs.github.io/exp-simulation-cartesian-system-dei/theory.html

**INPUTS:** X, Y, Z coordinates and axis moves.

**OUTPUTS:** Tool path and position of the printer head in the Cartesian system.